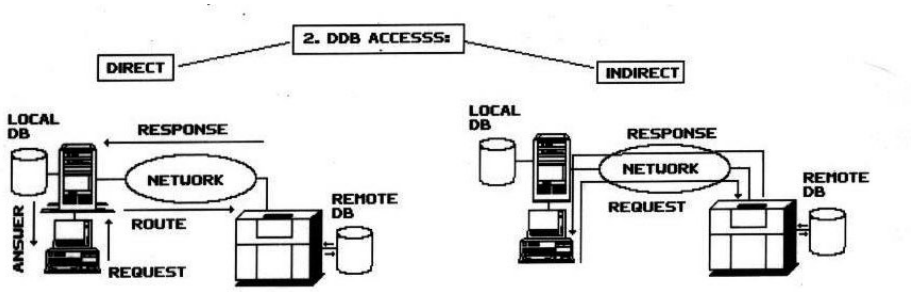
1. Define Distributed Database Systems and explain direct and indirect access methods

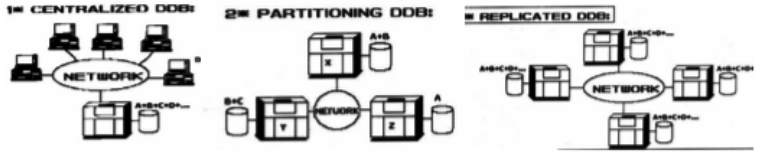
Distributed Database : is a collection of logically interrelated databases distributed over a network



Direct Access: when the requested data is in the local DB, and will not retrieved from the remote Network.

Indirect Access: When the requested data is not in the local DB and will be retrieved from the remote DB through the network, the request will use a route to the Remote DB, which the response will be through the network).

2. Compare between centralized, Partitioned, and replicated database processing systems, mention the main problem that faces the development of distributed data processing systems?



Centralized	Partitioned	Replicated
• One copy of the data resides at a central node. **Advantages :** • Centralized data control. • Clear standards. • Implementation. • Simple security. **Problems:** • High operating systems overhead. • Poor response time. • Increased reliability emphasis associated with dependence on a single system.	• Single copy of the data is split into parts and allocated geographically or functionally to specific nodes. • Queries must be decomposed and sub-requests sent to different nodes. • Used with relational databases in which the data is stored in two-dimensional tables called relations. These relations are composed of entities, which are either concrete or abstract objects, and attributes, which describe the objects	• Copies of key data file are stored at each site. • DDBMS must choose which copy or copies should be used in the request • For a query, any copy will satisfy the request, assuming all copies are updated through update synchronization. However, all copies must be the object of an update request.

3. Mention the main problem that faces the development of distributed data processing systems?

The main problem that faces the development of distributed data processing systems:

1) Data Recovery : Solution – Transaction management must track transaction to ensure the correct termination of the transaction.

2) Concurrency Control : Solution – Control access of several transactions on the same data

3) Deadlock /Lockup Handling : Solution –To prevent transactions from waiting each other's